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UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY
2022 INVITATIONAL A

Biology		Chemistry		Physics	
B01.	D	C01.	C	P01.	C
B02.	A	C02.	B	P02.	D
B03.	C	C03.	A	P03.	A
B04.	B	C04.	B	P04.	A
B05.	B	C05.	D	P05.	B
B06.	C	C06.	A	P06.	C
B07.	E	C07.	C	P07.	E
B08.	D	C08.	D	P08.	B
B09.	C	C09.	E	P09.	C
B10.	E	C10.	A	P10.	E
B11.	B	C11.	A	P11.	D
B12.	A	C12.	D	P12.	B
B13.	C	C13.	C	P13.	C
B14.	B	C14.	E	P14.	B
B15.	A	C15.	C	P15.	A
B16.	D	C16.	A	P16.	D
B17.	E	C17.	B	P17.	E
B18.	E	C18.	C	P18.	C
B19.	A	C19.	E	P19.	A
B20.	D	C20.	C	P20.	E

CHEMISTRY SOLUTIONS – UIL INVITATIONAL A 2022

C01. (C) 22.99 + 16.00 + 79.90 = 118.89 g/mol.

C02. (B) 100 g Al / 26.98 g/mol = 3.706 mol Al. 3.706 mol Al × 1 mol Al_2O_3/2 mol Al = 1.853 mol Al_2O_3.
1.853 mol Al_2O_3 × 101.96 g/mol = 189 g Al_2O_3.

C03. (A) $\nu\lambda = c$ and $E = h\nu$, so $E = hc/\lambda$. $E = (6.626 \times 10^{-34}\ \text{J·s})(3 \times 10^{8}\ \text{m/s})/(510 \times 10^{-9}\ \text{m}) = 3.90 \times 10^{-19}$ J

C04. (B) A central atom surrounded by four regions of electron density (a bond or a lone pair) will have tetrahedral electronic geometry. If there are no lone pairs, the molecular shape is the same as the electronic geometry.

C05. (D) Lattice energy depends on the ionic charges and the sizes of the ions. Higher ionic charges and smaller ionic radii result in higher lattice energies. Of these answer choices, MgO has the highest ionic charges (+2 ion and –2) and will have the strongest ion-ion attractions to overcome in order to melt the solid.

C06. (A) The two containers have the same temperature and the same number of moles, so the change in pressure depends only on the change in volume. $P_AV_A = P_BV_B$, $P_B = P_A(V_A/V_B) = 1.50(2.0/3.0) = 1.0$ atm

C07. (C) The problem states that the reaction is exothermic, so it is giving off heat and q is therefore negative. The reaction is producing a gas, so at constant pressure the volume is increasing. That means the system is doing work on the surroundings, therefore work is negative.

C08. (D). The heat of fusion for ice is 334 J/g and you have 454 grams, so 334 x 454 = 151,636 J = 152 kJ.

C09. (E): $K = \frac{[C][D]^2}{[A][B]^2}$ so $[D] = \sqrt{K\frac{[A][B]^2}{[C]}} = \sqrt{(1.5 \times 10^{-5}\frac{(0.0015)(0.0015)^2}{(0.0015)}} = 5.8 \times 10^{-6}$ M

C10. (A) KCl is the only neutral salt among the answer choices. KOH is a strong base and KNO_2 is a weak base and both would result in a solution with a pH greater than 7. HCl is a strong acid and HCN is a weak acid, so they would both result in a solution with a pH below 7.

C11. (A) $K_{sp} = [Ag^+][Cl^-]$, so $[Ag^+] = K_{sp}/[Cl^-] = 1.8 \times 10^{-10} / 5.5 \times 10^{-4}\ \text{M} = 3.3 \times 10^{-7}$ M

C12. (D) PV = nRT, so V = nRT/P. n = 12.8 mol, T = 298 K, R = 0.08206 L·atm/mol·K, and P = 1.00 atm.
V = (12.8)(0.08206)(298)/1.00 = 313 L total volume.

C13. (C). $E°_{cell} = E°_{reduction} - E°_{oxidation}$. Reduction occurs at the cathode and oxidation occurs at the anode, so $E°_{red} = +0.34$ V and $E°_{ox} = -1.66$ V. $E°_{cell} = (+0.34\ \text{V}) - (-1.66\ \text{V}) = +2.00$ V

C14. (E) A catalyst appears in the reaction mechanism first as a reactant and later as a product. There are no compounds in the reaction mechanism that fit this description.

C15. (C) Atomic radii decrease as you move up and to the right on the periodic table, so Cl would have the smallest radius of these atoms.

C16. (A) Oxidation states treat every compound as if it were ionic, even when we know it's not. When determining the oxidation states of atoms in compounds, assume H is +1 except in metal hydrides, and O is –2 except in peroxides. Halides are –1 unless they are the central atom, and N is –3 unless it is the central atom. The sum of the oxidation states of a neutral molecule adds up to 0.

C17. (B) The molar mass of PCl_5 = 1 × 30.97 + 5 × 35.45 = 208.22 g/mol.
100. g / 208.22 g/mol = 0.480 mol. 0.480 mol × 6.02×10^{23} molecules/mol = 2.89×10^{23} molecules.
2.89×10^{23} molecules × 6 atoms per molecule = 1.73×10^{24} atoms

C18. (C) After the 7*p* subshell comes the 8*s* subshell, so the next electron after the 7*p* subshell would go into the 8*s* orbital and the ground state electron configuration for element 119 would end in $8s^1$.

C19. (E) The vapor pressure of a mixture is $P = \chi_A P_A + \chi_B P_B$. Since pure A has a vapor pressure of 18 torr and pure B has a vapor pressure of 36 torr, the vapor pressure of any mixture of A and B will be somewhere in between, but will never be higher than the vapor pressure of pure B.

C20. (C) n=3 means the electron is from the third energy level, and $\ell = 0$ means he is from an *s*-orbital. Since he is the outermost electron in his atom, his home atom is one in which the outermost electron is in a 3*s* orbital. That could be either potassium or calcium, but calcium is not an answer choice so it must be potassium.

PHYSICS SOLUTIONS – UIL INVITATIONAL A 2022

P01. (C) page 15: "The concept of symmetry is simple, elegant, and intuitive. Moreover, throughout this book, we will see that symmetry is not just frivolous window dressing to a theory, but in fact is an essential feature that indicates some deep, underlying physical principle about the universe."

P02. (D) page 26: "In 1800, William Hershel had asked himself a simple question: What lies beyond the colors of the rainbow, which extend from red to violet? He took a prism, which created a rainbow in his lab, and placed a thermometer below the color red, where there was no color at all. Much to his surprise, the temperature of this blank area began to rise. In other words, there was a 'color' below red that was invisible to the naked eye but contained energy. It was called infrared light."

P03. (A) page 42: "In other words, acceleration in one frame is identical to the effect of gravity in another, which is due to space being curved."

P04. (A) Measurements of the spectra from sunspot regions indicate these regions are associated with very strong magnetic fields. These strong magnetic fields shut down the convection currents and lead to the cooler, darker spots that we see. Sunspots are generally associated with increased solar activity and strong solar winds. Sunspots are not associated with electric fields and do not affect the neutrino flux coming from the solar core.

P05. (B) First, we must do the subtraction in the parentheses. When subtracting, it is the value with the fewest number of decimal places that dictates the number of decimal places of the answer. Thus, when we subtract $19.13 - 16.2 = 2.93 = [2.9]3$; we have only two significant figures. The value 16.2 limits us to the tenths place, so the significant digits in the answer are also limited to the tenths place.

Now we can do the multiplication. We keep all of the digits from the subtraction because we don't round until the end. We do need to remember that the result of the subtraction only has two significant figures – this is indicated with brackets. So, we have $(6.18)([2.9]3) = 18.1074 = [18].1074$. With multiplication, the answer is limited to the number of significant figures of the input value with the least significant figures. The value [2.9]3 has only two significant figures, while 6.18 has three. Thus, the answer will be limited to two significant figures. Therefore, our answer to the correct number of significant figures is 18.

P06. (C) Since we know the time and the acceleration (9.8 m/s^2 downward), we should utilize the kinematic equation $y_f = y_i + v_{iy}t + \frac{1}{2}a_y t^2$. The final position is the forest floor ($y_f = 0$), and the initial position is the height at which the pinecone starts ($y_i = h$). Also, the pinecone starts from rest, so the initial velocity is zero ($v_{iy} = 0$). Putting it all together, we get $0 = h + 0 + (0.5)(-9.8)(3.5)^2$ which leads to $h = (4.9)(12.25) = 60$m.

P07. (E) The free body diagram for this situation includes three forces: weight, directed downward; the normal force, directed upward; and the applied force, directed to the right. There is no vertical motion, so the weight and the normal force balance one another. However, knowing that does not help solve the problem. In the absence of friction, the only horizontal force is the applied force, F. Utilizing Newton's acceleration law for the horizontal: $\sum F_x = F = ma_x \quad \rightarrow \quad 65.0N = (210kg)a_x$. This gives a horizontal acceleration of $a_x = 0.3095$ m/s^2.
Now we turn to kinematics. The box started from rest, so its initial velocity was zero. To find the final velocity, we go to the equation $v_{fx} = v_{ix} + a_x t \quad \rightarrow \quad v_{fx} = 0 + (0.3095)(2.5) = 0.774$ m/s.

P08. (B) To solve this problem, we rely on conservation of energy. Initially, the energy is stored in the compressed spring as elastic potential energy (EPE). After launching the block, the energy has been converted into kinetic energy (KE). Since there is no friction, we can set the initial elastic potential energy (EPE) equal to the final kinetic energy (KE).
Mathematically, $EPE = \frac{1}{2}kx^2 = (0.5)(390)(0.25)^2 = 12.19\,J = KE = \frac{1}{2}mv^2$.
Thus, we calculate $(0.5)(0.130)v^2 = 12.19 \;\; \rightarrow \;\; v^2 = 187.5 \;\; \rightarrow \;\; v = 13.7$m/s.

P09. (C) To solve this, we go to our angular kinematic equations. We don't know how long it took to complete one revolution, so we will use an equation without time in it: $\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$. The total angle through which the carousel went during acceleration was one revolution:
$\Delta\theta = 1rev = 2\pi$ radians. The initial angular velocity was zero, and we are given the final angular velocity. Putting it all together, we get:
$\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta = (2.30)^2 = 0 + 2\alpha(2\pi) \;\; \rightarrow \;\; 5.29 = 12.566\alpha \;\; \rightarrow \;\; \alpha = 0.42$ rad/s^2.

P10. (E) The oscillation period of a mass on a spring is given by $T = 2\pi\sqrt{\frac{m}{k}}$. Putting in the given values yields: $2.50 = 2\pi\sqrt{\frac{85}{k}}$. Solving for k gives: $\frac{2.50}{2\pi} = 0.398 = \sqrt{\frac{85}{k}} \;\; \rightarrow \;\; (0.398)^2 = 0.158 = \frac{85}{k}$. This gives us a spring constant of $k = \frac{85}{0.158} = 537 \approx 540$ N/m.

P11. (D) The heat transfer in this case is assumed to be entirely by conduction, so we will rely on the thermal conduction equation: $P = \frac{kA\Delta T}{L}$. We are given the thermal conductivity, k, and the thickness of the window, L. The window is rectangular, so the area of the window is given by:
$A = lw = (0.40m)(1.30m) = 0.52m^2$.
Finally, the temperature difference is $\Delta T = (22.0) - (-14.0) = 36.0$℃.
Putting this all into the thermal conduction equation, we get $P = \frac{(0.80)(0.52)(36.0)}{(0.0250)} = 599\,W \approx 600\,J/s$.
Here we recall that a Watt is a Joule per second, allowing us to conclude that this power equals the amount of heat energy lost through the window each second.

P12. (B) The two resistors are in series, so the total equivalent resistance is $R_{eq} = R_1 + R_2 = 810 + 560 = 1370\Omega$. Then, using Ohm's Law, we can find the current flowing in the circuit: $I_0 = \frac{V_0}{R_{eq}} = \frac{12.0V}{1370\Omega} = 0.00876A$. Since the two resistors are in series, this same current flows through both resistors. Now we can use Ohm's Law again to find the voltage across the 560Ω resistor: $V_2 = I_0R_2 = (0.00876A)(560\Omega) = 4.91V$.

P13. (C) This is a direct application of Coulomb's Law. The force on either charge due to the presence of the other charge is given by $F = \frac{kQ_1Q_2}{r^2}$. The distance between the charges is
$r = 20.0 - (-15.0) = 35.0cm = 0.350m$.
Thus, the force is $F = \frac{(8.99 \times 10^9)(9.50 \times 10^{-6})(12.5 \times 10^{-6})}{(0.350)^2} = 8.71\,N$.

P14. (B) The radius of the circle traced out by a charged particle in a magnetic field is given by the equation $r = \frac{mv}{qB}$. Solving for velocity in this equation yields: $v = \frac{rqB}{m}$. On the page of useful constants, we can find the charge and mass of a proton. Putting these in with the given radius and magnetic field gives us: $v = \frac{(0.560m)(1.602 \times 10^{-19}C)(0.60T)}{(1.67265 \times 10^{-27}kg)} = 3.22 \times 10^7$m/s. Although large, this is only about 1/10 of the speed of light, so we do not have to consider any relativistic effects.

P15. (A) This problem is a direct application of Faraday's Law. Using that law, we can find the voltage induced in the loop by the changing magnetic field: $\mathcal{E} = -\frac{\Delta\Phi_B}{\Delta t} = -\frac{\Delta(BA)}{\Delta t}$. The area of the loop does not change, and (in meters-squared) is equal to $A = (0.40m)^2 = 0.16m^2$. Since the magnetic field is perpendicular to the face of the loop, we don't need to worry about any angles. Therefore, we get: $\mathcal{E} = -\frac{A\Delta B}{\Delta t} = -\frac{0.16\,(B_f - B_i)}{\Delta t} = -0.16\frac{0\,T - 0.055\,T}{0.075\,sec} = 0.1173\,V$. This is the voltage induced in the loop. To get the current, we use Ohm's Law: $I = \frac{V}{R} = \frac{0.1173V}{1.50\Omega} = 0.0782A = 78.2mA$.

P16. (D) First, we need to find the image location of the magnified print. To do this, we use $\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$. In centimeters, this gives: $\frac{1}{3.50} + \frac{1}{q} = \frac{1}{5.00} \quad \rightarrow \quad q = -11.67cm$. This is the image location. Since it is negative, we know that it is a virtual image.
Now we can calculate the magnification: $M = -\frac{q}{p} = -\frac{-11.67}{3.50} = 3.33$.

P17. (E) The fact that this is a four-level laser system is cool, but all we really need to know is the energy difference between the two levels that produce the laser light. Those levels have energies of 2.21eV and 0.62eV. So, $\Delta E = 2.21 - 0.62 = 1.59\text{eV}$. Now we can determine the wavelength of the laser light: $\lambda = \frac{1240\,eVnm}{\Delta E} = \frac{1240}{1.59} = 780nm$.

P18. (C) Alpha particles carry away four units of mass and two units of charge for every particle. This means that the emission of seven alphas will reduce the atomic mass by $7 * 4 = 28$, and reduce the atomic number by $7 * 2 = 14$. Negative beta particles do not change the atomic mass but do increase the atomic number by one; so, emitting four beta particles will increase the atomic number by $1 * 4 = 4$. Gamma rays carry away excess energy, but do not change the atomic mass nor the atomic number.

We start with an atomic mass of 292. Taking into account all of the emitted particles, then this atomic mass with change to $292 - 28 - 0 - 0 = 264$. Similarly, the atomic number starts at 114. It will change to $114 - 14 + 4 - 0 = 104$. The element with atomic number 104 is Rutherfordium, so the atom left after these decays is Rutherfordium-264.

P19. (A) On a position-time graph, a straight line indicates that the speed of the runner is constant. Furthermore, the speed of the runner is equal to the slope of the line. A best-fit line is already given, so we just need to identify two points on that line and calculate the slope. In examining the line, one point is obvious: (30s, 300m). For the second point, I will choose (15s, 200m) which also looks pretty clear. Now to calculate the slope: $slope = \frac{\Delta y}{\Delta x} = \frac{300m - 200m}{30s - 15s} = \frac{100m}{15s} = 6.7m/s = speed$.

P20. (E) Although I don't recommend connecting vegetables into electrical circuits, this problem does give us the information we need to determine the resistance of the zucchini. Unsurprisingly, the zucchini is not entirely ohmic – a threshold voltage is required before any current will flow. If you made a plot of voltage versus current for this data, the threshold voltage would appear as a non-zero y-intercept. However, once the threshold voltage is exceeded, the plot is a straight line – and the resistance equals the slope of that line. In the case of tabulated data, we don't need to make the plot, we can estimate the slope based on any two of the data points given. I will use the first and last points: $slope = R = \frac{\Delta V}{\Delta I} = \frac{15.0V - 5.0V}{20.0\mu A - 1.33\mu A} = \frac{10.0V}{18.67 \times 10^{-6}A} = 5.35 \times 10^5\Omega = 535k\Omega$.
Note: you will get the same result using any pair of the data points given.